

Problem 1: Die Yield and Cost per Die

A 22 cm diameter wafer, A, costs \$30, contains 110 dies, and has a defect density of 0.025 defects/cm². Another wafer, B, has a 25 cm diameter, costs \$28, contains 130 dies, and has 0.018 defects/cm². Assume $N = 3.5$.

- Define 'yield' and explain its significance in semiconductor manufacturing.
- Explain why ' $\approx$ ' is used in the formula Dies per wafer $\approx$ Wafer area / Die area instead of '='.
- Compute the yield for both wafers.
- Compute the cost per die for both wafers.

Problem 2: Execution Time Comparison between Devices

Two laptops, Alpha and Beta, use variants of the same ISA. Alpha has a clock rate of 2.4 GHz and CPIs for instruction classes A–D as 6, 3, 2, 5. Beta has a clock rate of 3.2 GHz and CPIs of 4, 2, 2, 2.

A program with 2.0×10^6 instructions is made up of:

- 40% class A
- 30% class B
- 20% class C
- 10% class D

- Compute the average CPI for both Alpha and Beta.
- Compute the execution time for both systems.
- Given a reference system takes 90 ms to run the same program, compute the SPECRatio for Alpha.

Problem 3: Performance and Clock Rate Calculation

Three CPUs—C1, C2, and C3—execute the same instruction set.

- C1: 2.8 GHz, CPI = 1.2

- C2: 3.5 GHz, CPI = 1.8

- C3: 2.2 GHz, CPI = 0.9

- a. Identify which CPU has the highest performance in MIPS.
- b. If each takes 8 seconds to execute a program, calculate the number of clock cycles and instructions executed.
- c. If we reduce execution time by 25% and CPI increases by 15%, what clock rate is needed to maintain this improvement?

Problem 4: Instruction Execution Time Reduction

A system executes a program in 300s, with time breakdown as:

- 100s on multiply instructions

- 120s on divide instructions

- 80s on logical AND instructions

- a. What is the new total time if multiply operations are improved by 25%?
- b. Can a 20% total time reduction be achieved by optimizing only logical AND instructions? Show calculation.

Problem 5: Comparing MIPS and Real Performance

Processor A: Clock = 5 GHz, CPI = 1.0, Instruction count = $8.0E9$

Processor B: Clock = 3.2 GHz, CPI = 0.7, Instruction count = $4.0E9$

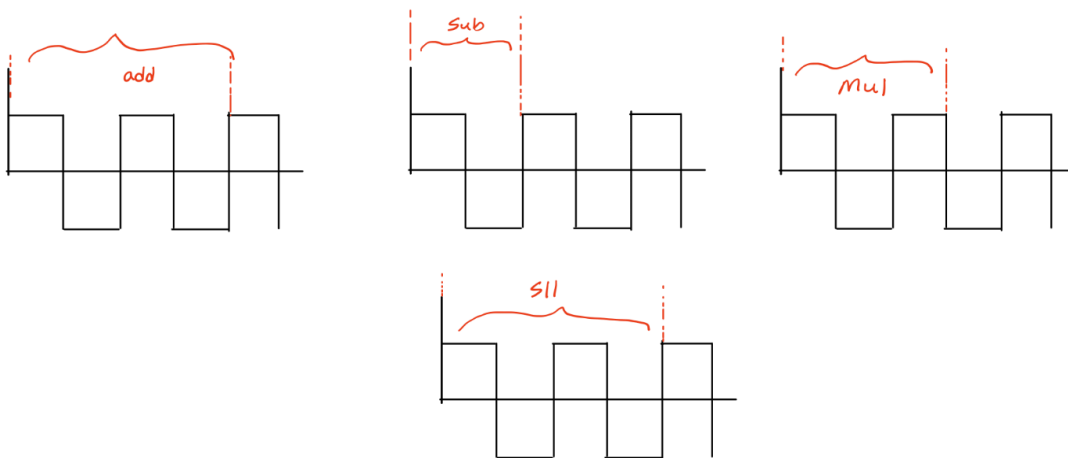
- Compute MIPS for both processors.
- Determine which processor is faster in execution time.
- Does MIPS reflect real performance here? Explain why or why not.

Problem 6: Power Consumption Trends

A modern processor design reduces its capacitance by 10%, and runs at 85% of the original voltage and 90% of the original frequency.

- Calculate the percentage reduction in dynamic power consumption.
- If the original power was 100W, what is the new power usage?
- Why is reducing voltage more impactful than reducing frequency?

Problem 7: Clock Cycle Graph Based Problems



Program **X** is divided into four classes according to their **CPI** (Add,Sub,Mul,SLL). The **instruction counts** are 21,3,15,10 respectively. Reference for **program X** is

1080ps. The period of each of the instruction clocks in the given reference image is 50ps.

Now, answer the following questions,

- a) What is the frequency?
- b) What is the CPI for Add,Sub,Mul,SLL?[Hint: CPI can be fraction numbers]
- c) What is Avg. CPI?
- d) Find out the execution time of the program?
- e) Find the SPEC ratio?
- f) If you want to improve the performance by 1.2 times, what improvement do you need to include in the program's add operation?